STRENGTH OF MATERIALS

Subject Code : 23A01302T

UNIT – III

Flexural and Shear Stresses

Handwritten Study Notes

Syllabus topics covered in this unit :

Theory of simple bending — Assumptions — Bending equation — Neutral axis — Bending stresses — Section modulus of rectangular, circular (solid & hollow), I, T, Angle and Channel sections — Design of simple beams — Shear stress formula and distribution (rectangular, circular, I, T, Angle) — Torsion of circular shafts

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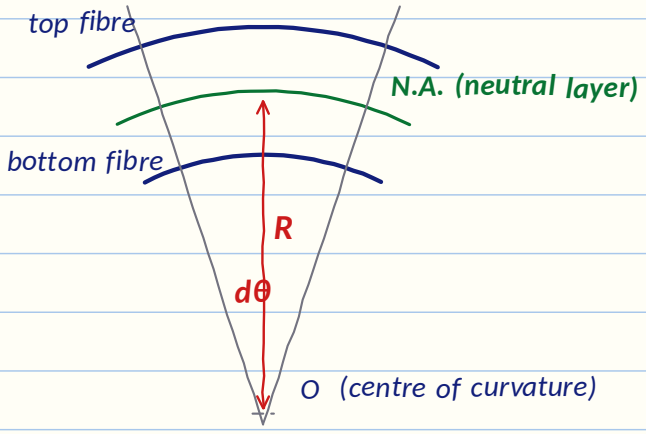
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→ 3.2 Section Modulus of Common Sections
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→ 3.4 Shear Stresses in Beams
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3.1 Theory of Simple Bending

Pure (simple) bending : a beam segment subjected to a constant bending moment with zero shear force – only bending stresses (σ) act in it.

Assumptions

- Beam material is homogeneous and isotropic ; obeys Hooke's law (stress within elastic limit).
- E is same in tension and compression.
- Plane sections remain plane after bending (transverse sections stay ⊥ to the neutral surface).
- Beam is initially straight, of uniform section ; radius of curvature is large compared with depth.
- Loads act in the plane of symmetry (no twisting) ; each layer is free to expand/contract.
- Shear effects are neglected in pure bending.



Layer at distance y from N.A.:

original length = $R\,d\theta$

new length = $(R + y)\,d\theta$

strain $\varepsilon = y\,d\theta / (R\,d\theta) = y / R$

stress $\sigma = E\,\varepsilon = E\,y / R$

$\Rightarrow \quad \sigma / y = E / R$

Neutral axis (N.A.)

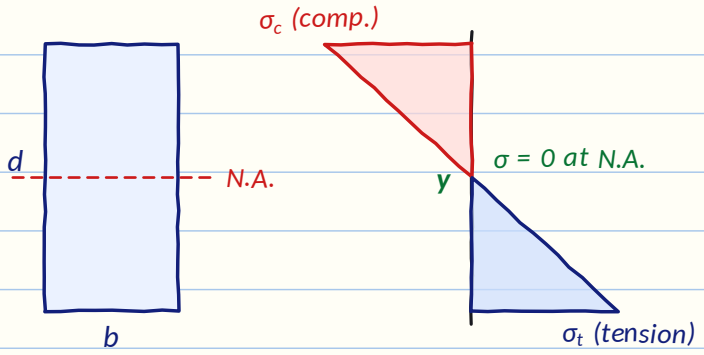
The layer that is neither stretched nor compressed is the neutral layer ; its trace on a cross-section is the neutral axis. N.A. passes through the centroid of the section.

- Proof : net axial force = $\int \sigma\,dA = (E/R) \int y\,dA = 0 \Rightarrow \int y\,dA = 0$ (centroidal axis).

Derivation of Bending Equation

- Layer at distance y from N.A. : strain $\varepsilon = y / R \Rightarrow \sigma = E\,y / R$.
- Moment of the force on strip dA about N.A. : $dM = \sigma\,dA \cdot y = (E/R)\,y^2\,dA$.
- Total $M = (E/R) \int y^2\,dA = (E/R)\,I$ where I = moment of inertia about N.A.

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$



σ varies linearly with y

N.A. passes through centroid

$\sigma_{max} = M\, y_{max} / I = M / Z$

$Z = I / y_{max}$ (section modulus)

Moment of resistance M = σ Z

- Bending stress varies linearly ; zero at N.A. ; max at the extreme fibre.

$$\sigma_{max} = \frac{M}{I}\, y_{max} = \frac{M}{Z} \qquad Z = \frac{I}{y_{max}}$$

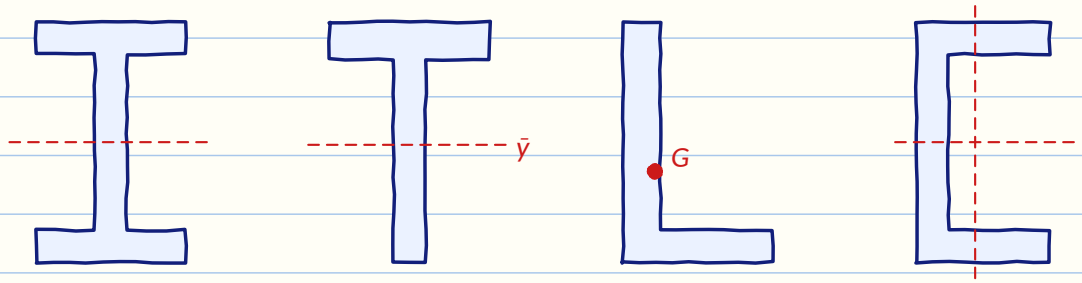
- Z = section modulus (mm^3) : a measure of the strength of the section in bending. Moment of resistance $M_r = \sigma_{permissible} \times Z$.

3.2 Section Modulus of Common Sections

- Parallel axis theorem : $I = I_G + A\,h^2$ (h = distance between parallel axes).
- Perpendicular axis theorem : $I_z = I_x + I_y$ (plane areas).

Section	I (about N.A.)	$Z = I / y_{max}$
Rectangle $b \times d$	$b\,d^3/12$	$b\,d^2/6$
Hollow rectangle (B×D, b×d)	$(B\,D^3 - b\,d^3)/12$	$(B\,D^3 - b\,d^3)/6D$
Solid circle dia. d	$\pi\,d^4/64$	$\pi\,d^3/32$
Hollow circle D, d	$\pi\,(D^4 - d^4)/64$	$\pi\,(D^4 - d^4)/32\,D$
Triangle (base b, ht h) about base	$b\,h^3/12$	—
Triangle about centroid	$b\,h^3/36$	$b\,h^2/24$ (apex side)

- For a given area, I and Z are larger when material is placed away from the N.A.
  ⇒ I, T, hollow sections are economical.



I-section T-section Angle Channel

Procedure for I, T, Angle, Channel

- 1. Split the section into simple rectangles : areas A_1, A_2, ...
- 2. Find centroid : $\bar{y} = (\sum A\,y) / \sum A$ (measure from one edge) ; similarly $\bar{x}$ for unsymmetrical sections.
- 3. $I_{NA} = \sum (I_G + A\,h^2)$ of each rectangle about the centroidal axis.
- 4. $Z_{top} = I / y_{top}$, $Z_{bottom} = I / y_{bottom}$; use the smaller Z for max stress.

Example 1 (T-section)

Flange 100 × 20 mm ; web 20 × 100 mm (overall depth 120 mm). Find $\bar{y}$, I and Z.

$A_1 = 2000$ ($y_1 = 10$ from top), $A_2 = 2000$ ($y_2 = 70$) ⇒ $\bar{y} = (2000\times10 + 2000\times70)/4000 =$ **40 mm from top**

$I = (100\times20^3/12 + 2000\times30^2) + (20\times100^3/12 + 2000\times30^2)$

$I = 66\,667 + 1.8\times10^6 + 1.667\times10^6 + 1.8\times10^6 =$ **5.333×10^6 mm^4**

$Z_{top} = I/40 =$ **1.333×10^5 mm^3** ; $Z_{bottom} = I/80 =$ **0.667×10^5 mm^3**

Example 2 (I-section)

Flanges 200 × 20 ; web 10 × 300 ; overall depth 340 mm.

$I = [200 \times 340^3 - (200 - 10) \times 300^3] / 12 = [7.8608\times10^9 - 5.13\times10^9]/12 =$ **2.2757×10^8 mm^4**

$Z = I / 170 =$ **1.3386×10^6 mm^3**

Example 3 (Angle 100 × 80 × 10)

$A_1 = 100\times10 = 1000$ $(x=5, y=50)$; $A_2 = 70\times10 = 700$ $(x=45, y=5)$

$\bar{x} = (5000 + 31500)/1700 = 21.47$ mm ; $\bar{y} = (50000 + 3500)/1700 = 31.47$ mm

$I_{xx} = (10\times100^3/12 + 1000\times18.53^2) + (70\times10^3/12 + 700\times26.47^2) \approx$ **1.673×10^6 mm^4**

Channel section

- Symmetrical about the horizontal axis : I_{xx} by the usual formula ; unsymmetrical about the vertical axis ⇒ find $\bar{x}$ from the back of the web ; $I_{yy} = \sum(I_G + A\,h^2)$ about $\bar{x}$.
- Loads should act through the shear centre of a channel, otherwise twisting occurs.

3.3 Bending Stress Problems and Design

Example 4

Rectangular beam 100 mm × 200 mm, simply supported over 4 m, U.D.L. 10 kN/m.

$M_{max} = w L^2/8 = 10 \times 16/8 = 20$ kN·m $= 20\times10^6$ N·mm

$I = 100 \times 200^3/12 = 66.67\times10^6$ mm^4

$\sigma_{max} = M\,y/I = 20\times10^6 \times 100 / 66.67\times10^6 =$ **30 N/mm^2**

Example 5 (Circular section)

Solid circular beam $d = 100$ mm, $M = 5$ kN·m.

$Z = \pi d^3/32 = 98\,175$ mm$^3 \Rightarrow \sigma = 5\times10^6 / 98\,175 =$ **50.9 N/mm^2**

Example 6 (Cantilever – extreme stresses)

Cantilever 2 m, point load 5 kN at the end, T-section of Example 1 with flange on top.

$M_A = 5 \times 2 = 10$ kN·m (hogging ⇒ top in tension)

σ_{top} (tension) $= M/Z_{top} = 10\times10^6 / 1.333\times10^5 =$ **75 N/mm^2**

σ_{bottom} (compression) $= 10\times10^6 / 0.667\times10^5 =$ **150 N/mm^2**

Design of simple beams

Procedure : (1) find M_{max} from the B.M.D. (2) $Z_{required} = M_{max} / \sigma_{permissible}$ (3) choose a section with $Z \geq Z_{req}$ (4) check shear stress and deflection.

Example 7 (Rectangular timber beam, depth = 2 × width)

S.S. beam 5 m, total load 12 kN/m (incl. self weight), $\sigma_{perm} = 150$ N/mm^2.

$M = 12 \times 25/8 = 37.5$ kN·m ; $Z_{req} = 37.5\times10^6/150 = 2.5\times10^5$ mm^3

$Z = b\,d^2/6 = b\,(2b)^2/6 = 2b^3/3 = 2.5\times10^5 \Rightarrow b = 72.1$ mm

Provide **75 mm × 150 mm** ($Z = 2.81\times10^5$ mm^3 > required ✓)

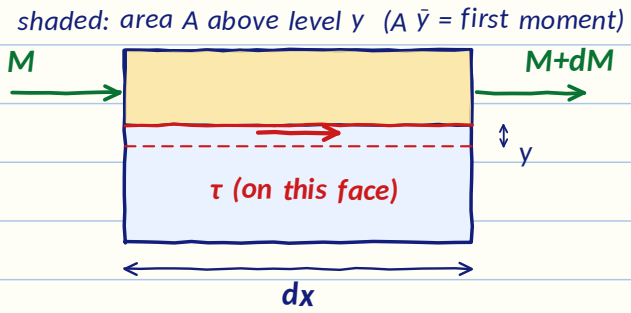
- For steel use I-sections : select the lightest rolled section from tables with $Z_{xx} \geq 250$ cm^3.

Strength comparison

- Beams of equal area : I-section ≫ square > circle for the same M (larger Z).
- Rectangle on edge vs flat : Z ratio = d/b (deep side vertical is stronger).

3.4 Shear Stresses in Beams

- A shear force F induces vertical shear stress τ on the cross-section and equal horizontal (complementary) shear stress on longitudinal planes.



Force on left face = $\int \sigma\, dA = (M/I)\, A\, \bar{y}$

Force on right face = $((M+dM)/I)\, A\, \bar{y}$

Difference = $(dM/I)\, A\, \bar{y}$

Balanced by $\tau\, b\, dx$:

$\tau\, b\, dx = (dM/I)\, A\, \bar{y}$

$\tau = F\, A\, \bar{y} / (I\, b)$ $(F = dM/dx)$

$$\tau = \frac{F\, A\, \bar{y}}{I\, b}$$

- F = shear force ; A = area of the part of section beyond the level considered ; $\bar{y}$ = distance of its centroid from N.A. ; b = width at that level ; I = M.I. of whole section about N.A.

Rectangular section (b × d)

- At height y from N.A. : $A\, \bar{y} = b\,(d/2 - y)(d/2 + y)/2 \Rightarrow \tau = F/2I((d^2)/4 - y^2)$ (parabolic).

$$\tau_{max}\ (\text{at N.A.}) = \frac{3F}{2\, b\, d} = 1.5\, \tau_{avg} \qquad \tau = 0 \text{ at top/bottom}$$

Circular section (radius R)

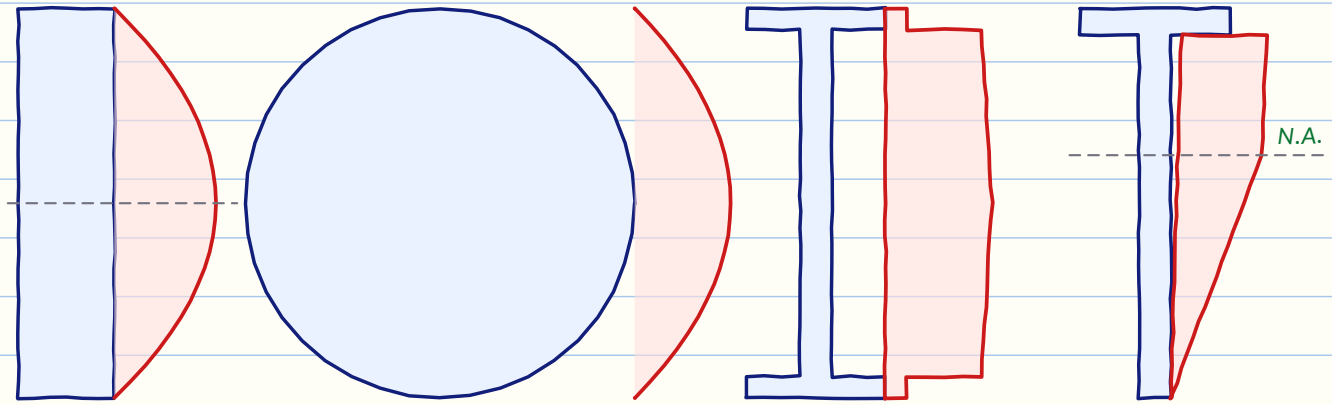
$$\tau_{max} = \frac{4F}{3\,\pi\, R^2} = \frac{4}{3}\, \tau_{avg} \quad (\text{at N.A.})$$

I-section

- Flange : τ small (wide b) ; at flange–web junction τ jumps (b changes from flange width to web thickness) ; max at N.A.
- Web carries about 95 % of the shear force, flanges carry bending ⇒ economical.

T-section and Angle

- T : same trend ; max at N.A. (not at the middle of depth) ; zero at free edges. Angle : max at N.A. of the leg ; shear centre lies at the corner.



Rectangle $\tau_{max} = 1.5\ \tau_{avg}$ | **Circle** $\tau_{max} = (4/3)\ \tau_{avg}$ | **I-section** web carries ≈ 95 % of F | **T-section** max at N.A.

Example 8 (Rectangular)

$b = 100$, $d = 200$ mm, $F = 30$ kN : $\tau_{avg} = 30000/20000 = 1.5 \Rightarrow \tau_{max} = 2.25$ N/mm^2

Example 9 (I-section of Example 2, F = 100 kN)

$A\bar{y}$ at N.A. $= 200\times20\times160 + 10\times150\times75 = 7.525\times10^5$ mm^3

$\tau_{NA} = 100000 \times 7.525\times10^5 / (2.2757\times10^8 \times 10) =$ **33.07 N/mm^2**

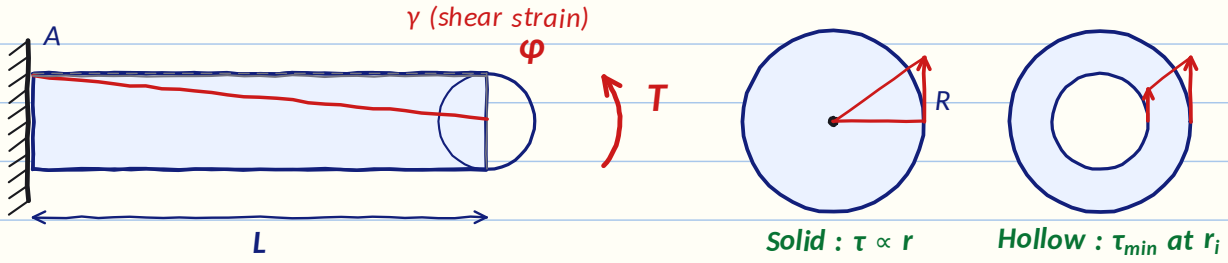
At web (junction) : $A\bar{y} = 6.4\times10^5 \Rightarrow \tau =$ **28.1 N/mm^2** ; in flange ($b = 200$) : $\tau =$ **1.41 N/mm^2**

3.5 Torsion of Circular Shafts

Torsion : a shaft is twisted by a torque T about its axis ⇒ shear stress on cross-sections.

Assumptions

- Material is homogeneous, isotropic, elastic (Hooke's law) ; shaft is circular and uniform.
- Plane cross-sections remain plane and radii remain straight after twisting.
- Twist is uniform along the length ; stress is within proportional limit.



Derivation of torsion equation

- Shear strain at radius r : $\gamma = r\,\varphi / L$; $\tau = G\,\gamma = G\,r\,\varphi / L$.
- Torque on ring dA at radius r : $dT = \tau\, r\, dA = (G\,\varphi / L)\, r^2\, dA \Rightarrow T = (G\,\varphi / L)\, J$.

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\,\varphi}{L}$$

- J = polar moment of inertia ; τ_{max} at outer surface (r = R).

Shaft	J	$Z_p = J/R$
Solid dia. d	$\pi\, d^4/32$	$\pi\, d^3/16$
Hollow D, d	$\pi\,(D^4 - d^4)/32$	$\pi\,(D^4 - d^4)/16\,D$

- Torsional rigidity = G J ; angle of twist $\varphi = T\,L / G\,J$ (radian).

$$\text{Power } P = \frac{2\,\pi\,N\,T}{60} \quad (T \text{ in N·m, } P \text{ in watt})$$

- Hollow shafts are stronger and lighter per kg than solid shafts (material near the axis is under-stressed).

Example 10 (Solid shaft)

$d = 50$ mm, $\tau_{perm} = 40$ N/mm^2, $N = 300$ rpm, $G = 80$ GPa, $L = 2$ m.

$T = \tau \pi d^3/16 = 40 \times 24\,544 = 9.817\times10^5$ N·mm = **0.982 kN·m**

$P = 2\pi N T / 60 = 2\pi \times 300 \times 981.7 / 60 =$ **30.84 kW**

$\varphi = T L / G J = 9.817\times10^5\times2000 / (8\times10^4 \times 6.136\times10^6) =$ **0.0040 rad = 0.229°**

Example 11 (Hollow shaft D = 60, d = 40 mm, T = 1 kN·m)

$J = \pi (60^4 - 40^4)/32 = 1.021\times10^6$ mm^4

$\tau_{max} = T R / J = 10^6 \times 30 / 1.021\times10^6 =$ **29.4 N/mm^2**

3.6 Derivations and More Solved Problems

Moment of inertia – by integration

- Rectangle ($b \times d$) about N.A. : strip of width b, thickness dy at distance y :

$$I = \int y^2\, b\, dy \quad (\text{from } -d/2 \text{ to } d/2) = \frac{b\,d^3}{12}$$

- Circle (radius R) : polar M.I. $J = \int r^2 (2\pi r\, dr) = (\pi R^4)/2 = (\pi d^4)/32$.
- By the perpendicular axis theorem $I_x = I_y = J/2 \Rightarrow$ **$I = \pi d^4/64$**.

Example 12 (Channel section)

Channel : web 10 × 200 mm, two flanges projecting 70 × 10 mm each (flange width 80, overall depth 200).

$A = 2000 + 2 \times 700 = 3400$ mm^2 ; $\bar{x} = (2000\times5 + 1400\times45)/3400 =$ **21.47 mm** from back of web

$I_{xx} = 10\times200^3/12 + 2\,[70\times10^3/12 + 700\times95^2] =$ **1.931×10^7 mm^4** $\Rightarrow Z_{xx} = 1.931\times10^5$ mm^3

$I_{yy} = [200\times10^3/12 + 2000\times16.47^2] + 2\,[10\times70^3/12 + 700\times23.53^2] \approx$ **1.906×10^6 mm^4**

Example 13 (Stress distribution in I-section, M = 100 kN·m)

Use the I-section of Example 2 ($I = 2.2757\times10^8$ mm^4, $Z = 1.3386\times10^6$ mm^3).

$\sigma_{max} = M/Z = 100\times10^6 / 1.3386\times10^6 =$ **74.7 N/mm^2** (top/bottom fibre)

At web–flange junction ($y = 150$) : $\sigma = 74.7 \times 150/170 =$ **65.9 N/mm^2**

Flanges alone : $I_f = (200\times340^3 - 200\times300^3)/12 = 2.05\times10^8 \Rightarrow$ about 90 % of I – so flanges resist most of the bending moment.

Example 14 (Cantilever with U.D.L. – check of rectangular section)

Cantilever 1.5 m, U.D.L. 20 kN/m, section 120 mm × 240 mm.

$M_{max} = 20 \times 1.5^2/2 = 22.5$ kN·m ; $Z = 120 \times 240^2/6 = 1.152\times10^6$ mm^3

$\sigma_{max} = 22.5\times10^6 / 1.152\times10^6 =$ **19.53 N/mm^2** ; $F_{max} = 30$ kN $\Rightarrow \tau_{max} = 1.5 \times$

30000/28800 = **1.56 N/mm²**

Example 15 (Shear stress in T-section, F = 20 kN)

Section of Example 1 : $I = 5.333\times10^6$ mm⁴, N.A. 40 mm from top.

At N.A. ($b = 20$) : $A\bar{y} = 2000\times30 + 20\times20\times10 = 64\,000$ mm³ ⇒ $\tau =$ $20000\times64000/(5.333\times10^6\times20)$ = **12.0 N/mm²**

Junction – web side ($b = 20$) : $A\bar{y} = 60\,000$ ⇒ **11.25 N/mm²** ; flange side ($b = 100$) : **2.25 N/mm²**

Hollow vs solid shaft (same torque, same τ_{max})

Solid $d = 100$ mm ; hollow shaft with $d_i / D_o = 0.6$.

$D_o^3 (1 - 0.6^4) = 100^3 \Rightarrow D_o = 104.7$ mm ; $d_i = 62.8$ mm

$A_{hollow} = \pi/4\,(104.7^2 - 62.8^2) = 5512$ mm² vs $A_{solid} = 7854$ mm² ⇒ **≈ 30 % saving in weight**

Example 16 (Safe U.D.L. on a timber beam)

S.S. beam 6 m, section 150 × 300 mm, $\sigma_{perm} = 10$ N/mm².

$Z = 150 \times 300^2/6 = 2.25\times10^6$ mm³ ; $M_r = \sigma Z = 22.5$ kN·m

$w L^2/8 = 22.5 \Rightarrow w = 22.5 \times 8 / 36 =$ **5 kN/m** (safe U.D.L., incl. self weight)

Example 17 (Hollow circular beam)

$D = 120$ mm, $d = 80$ mm, $M = 8$ kN·m.

$I = \pi\,(120^4 - 80^4)/64 = 8.168\times10^6$ mm⁴

$\sigma_{max} = M y/I = 8\times10^6 \times 60 / 8.168\times10^6 =$ **58.8 N/mm²**

Radius of curvature $R = E I / M = 2\times10^5 \times 8.168\times10^6 / 8\times10^6 =$ **204 m**

Example 18 (Diameter of a circular beam)

$M = 12$ kN·m, $\sigma_{perm} = 100$ N/mm².

$Z = M/\sigma = 1.2\times10^5$ mm³ $= \pi d^3/32 \Rightarrow d^3 = 1.222\times10^6 \Rightarrow$ **d = 107 mm → provide 110 mm**

Beam of uniform strength (concept)

- Section varies so that σ_{max} is the same at every section : M / Z = constant.
- Cantilever with end load : rectangular b constant ⇒ $d \propto \sqrt{x}$ (parabolic profile) ; d constant ⇒ $b \propto x$ (triangular plan).
- Saves material ; used in leaf springs.

Common mistakes to avoid

- Using the wrong y : y is always measured from the neutral axis (centroid), not from the bottom edge.
- For unsymmetrical sections compute stresses at BOTH extreme fibres : σ_t and σ_c

are different.

- Keep units consistent : N and mm (M in N·mm = kN·m × 10^6).

Shafts in series and parallel

- Series (different sections, same torque T) : total twist $\varphi = \varphi_1 + \varphi_2 = T\,L_1/G_{1J1} + T\,L_2/G_{2J2}$.
- Parallel (composite shaft, same twist) : $\varphi_1 = \varphi_2$ and $T = T_1 + T_2 \Rightarrow T_1/T_2 = G_1\,J_1 / G_2\,J_2$ (for equal lengths).

Formula summary – Unit III

Quantity	Formula
Bending equation	$M/I = \sigma/y = E/R$
Section modulus	$Z = I / y_{max}$
Shear stress	$\tau = F\,A\,\bar{y}\,/(I\,b)$
Rectangle τ_{max}	$3F/2bd = 1.5\,\tau_{avg}$
Circle τ_{max}	$4F/3A = 1.33\,\tau_{avg}$
Torsion	$T/J = \tau/r = G\,\varphi/L$
Power transmitted	$P = 2\pi\,N\,T\,/\,60$

Short questions

- Define section modulus. A: $Z = I / y_{max}$ (mm^3).
- Where is bending stress zero? A: at the neutral axis ; shear stress is zero at extreme fibres.
- Max shear stress / average : rectangle 1.5 ; solid circle 4/3.
- What is polar modulus? A: $Z_p = J / R$.